A First Course in Abstract Algebra

Section 2: Binary Operations

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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2 Binary Operations

Exercise 2.1: Reading the operation table

Use the first factor's row and the second factor's column. Table 2.26 gives $b * d = e$ and $c * c = b$. For the nested expression, work from the innermost parentheses: $a * c = c$, then $c * e = a$, then $a * a = a$. Thus $[(a * c) * e] * a = a$.

Exercise 2.2: A test that happens to agree

The table gives $(a * b) * c = b * c = a$ and $a * (b * c) = a * a = a$. These values agree, but one successful triple does not establish associativity: that law must hold for every ordered triple. This computation alone therefore cannot decide whether the operation is associative.

Exercise 2.3: A failed associativity test

Since $b * d = e$, $(b * d) * c = e * c = a$. On the other hand $d * c = b$, so $b * (d * c) = b * b = c$. The elements a, c differ, so this one triple disproves associativity.

Exercise 2.4: Testing commutativity

The operation is not commutative: $a * e = d$ whereas $e * a = b$. A commutative table must be symmetric across its main diagonal; these two off-diagonal entries already violate that requirement.

Exercise 2.5: A symmetric completion

Reflect each known entry across the diagonal, leaving the given diagonal entries unchanged. This forces

$*$	a	b	c	d
a	a	b	c	d
b	b	d	a	c
c	c	a	d	b
d	d	c	b	a

For instance $c * b = a$ forces $b * c = a$, and $b * d = c$ forces $d * b = c$. Every resulting entry is in the specified set and the table is symmetric, as required.

Exercise 2.6: An associative completion

The known entry $c * b = d$ lets us compute every missing product: associativity requires $d * x = (c * b) * x = c * (b * x)$. For $x = a, b, c, d$, the given row of b is b, a, c, d; applying the given row of c gives d, c, c, d. Thus

$*$	a	b	c	d
a	a	b	c	d
b	b	a	c	d
c	c	d	c	d
d	d	c	c	d

The question permits assuming an associative completion exists. The preceding equations show it must be this one; direct checking of its 4^3 triples confirms it is associative.

Exercise 2.7: Subtraction on the integers

This is neither commutative nor associative. For commutativity, $1 * 2 = -1$ while $2 * 1 = 1$. For associativity, $(1 * 2) * 3 = -1 - 3 = -4$, whereas $1 * (2 * 3) = 1 - (-1) = 2$. Every output is still an integer, so it is a binary operation despite these failed laws.

Exercise 2.8: Multiplication followed by adding one

Commutativity holds because $ab + 1 = ba + 1$. However

$$(a * b) * c = (ab + 1)c + 1 = abc + c + 1, \qquad a * (b * c) = a(bc + 1) + 1 = abc + a + 1.$$

Taking $a = 0, b = 0, c = 1$ gives 2 versus 1, so associativity fails.

Exercise 2.9: A scaled product

Commutativity follows from $ab/2 = ba/2$. For every rational a, b, c,

$$(a * b) * c = ((ab)/2)c/2 = abc/4 = a((bc)/2)/2 = a * (b * c).$$

Thus the operation is both commutative and associative. Products and division by 2 keep all outputs rational.

Exercise 2.10: Two raised to a product

The operation is commutative since $ab = ba$. It is not associative: with $(a, b, c) = (1, 1, 3)$,

$$(1 * 1) * 3 = 2 * 3 = 2^6 = 64, \qquad 1 * (1 * 3) = 1 * 8 = 2^8 = 256.$$

All outputs are positive integers, so closure is not an issue.

Exercise 2.11: Exponentiation

This operation is not commutative: $2^3 = 8 \neq 9 = 3^2$. It is not associative either:

$$(2 * 2) * 3 = 4^3 = 64, \qquad 2 * (2 * 3) = 2^8 = 256.$$

The distinction is between $(a^b)^c = a^{bc}$ and a^{b^c}, not a failure of closure.

Exercise 2.12: Counting all binary operations

For a set of size n, there are n^2 ordered input pairs and independently n choices for each output. Thus there are n^{n^2} operations. For one, two, and three elements this gives respectively 1, $2^4 = 16$, and $3^9 = 19683$. The table is an arbitrary function from $S \times S$ into S.

Exercise 2.13: Counting commutative operations

Only the diagonal and one half of the off-diagonal entries can be chosen freely. There are $n + n(n-1)/2 = n(n+1)/2$ such positions, each with n possible outputs, so the answer is $n^{n(n+1)/2}$. It is $2^3 = 8$ for two elements and $3^6 = 729$ for three elements. The remaining entries are forced by symmetry.

Exercise 2.14: A precise commutativity definition

A binary operation $*$ on a set S is commutative if and only if $a * b = b * a$ *for all* $a, b \in S$. The universal quantifier and the underlying set must be specified: agreement for just one pair does not suffice.

Exercise 2.15: An acceptable associativity definition

The given definition is correct: $(b * c) * a = b * (c * a)$ for all $a, b, c \in S$ is the associative law with the dummy variables renamed. For the more usual arrangement, one may write $(x * y) * z = x * (y * z)$ for all $x, y, z \in S$. No correction of its mathematical content is needed.

Exercise 2.16: Closure of a subset

A subset $H \subseteq S$ is closed under $*$ if and only if $a * b \in H$ for all $a, b \in H$. The proposed version incorrectly quantifies over all of S. Closure only asks about two inputs already in H.

Exercise 2.17: A subtraction rule on positive integers

The numerical output $a - b$ is uniquely determined for every input, so Condition 1 holds. Condition 2, closure, fails: $1 * 1 = 0 \notin \mathbb{Z}^+$ (and $1 * 2 = -1$ is another failure). Thus this does not define a binary operation on $\mathbb{Z}^+$.

Exercise 2.18: An exponentiation rule on positive integers

For each positive integer pair a, b, the integer a^b exists uniquely and is positive. Both uniqueness/everywhere-definition (Condition 1) and closure (Condition 2) hold. Hence this is a binary operation on the given set.

Exercise 2.19: A subtraction rule on all real numbers

For any two real numbers a, b, the difference $a - b$ is one uniquely determined real number. Therefore Conditions 1 and 2 both hold; subtraction is a binary operation on $\mathbb{R}$. Associativity is not required in the definition of binary operation.

Exercise 2.20: The least integer above two inputs

For positive integers a, b, the specified integer is exactly $\max(a, b) + 1$. It exists, is unique, and is positive. Thus both conditions hold and the rule defines a binary operation.

Exercise 2.21: An ambiguous lower bound

The condition permits every integer $c \geq a + b + 5$. For example, when $a = b = 1$, both 7 and 8 are allowed. Thus Condition 1 fails because no unique output is specified. All allowed outputs are positive, so there is no violation of Condition 2. The rule is not a binary operation.

Exercise 2.22: The largest integer below a product

Since ab is an integer, the largest integer strictly below it is uniquely $ab - 1$. Thus Condition 1 holds. But $1 * 1 = 0$ is not positive, so Condition 2 fails and the rule is not a binary operation on $\mathbb{Z}^+$.

Exercise 2.23: A matrix-shaped subset

Write $M(a, b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. For arbitrary real a, b, c, d,

$$M(a, b) + M(c, d) = M(a + c, b + d),$$

so **(a)** it is closed under addition. Also

$$M(a, b)M(c, d) = \begin{pmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{pmatrix} = M(ac - bd, ad + bc),$$

so **(b)** it is closed under multiplication. Exhibiting the same form proves closure for all inputs.

Exercise 2.24: Ten statements about operations

(a) False: addition on $\mathbb{Z}$ has $1 + 1 \neq 1$.

(b) True: treat $b * c \in S$ as one element and apply commutativity to it and a.

(c) False: matrix multiplication is associative but generally noncommutative; specifically take $a = E_{12}$, $b = E_{21}$, $c = I_2$. Then $a(bc) = E_{11}$ but $(bc)a = E_{22}$.

(d) False: function composition, set union, and matrix multiplication are important operations on nonnumeric objects.

(e) False: the equation must hold for every pair. Subtraction has $0-0 = 0-0$ but is not commutative.

(f) True: on $\{s\}$ every product must be s, so both laws hold.

(g) True: exactly one output in particular means at least one.

(h) True: exactly one output in particular means at most one.

(i) True: this is the defining requirement, together with the specified codomain.

(j) False: multiple allowed outputs contradict being a function.

Exercise 2.25: Two operations on a nonnumeric set

Take S to be the set of all finite words in the alphabet $\{A, B\}$, including the empty word. Define $u * v$ to be concatenation and $u *' v = u$. Each ordered pair has exactly one output in S. They are different: for $u = A, v = B$, concatenation gives AB, while the second operation gives A. The alphabet and finiteness requirement make the set unambiguous.

Exercise 2.26: Rearranging four factors

Use only the stated laws, one step at a time:

$$\begin{aligned}(a * b) * (c * d) &= (c * d) * (a * b) && \text{(commutativity)}\\ &= ((c * d) * a) * b && \text{(associativity on } c * d, a, b)\\ &= ((d * c) * a) * b && \text{(commutativity on } c, d).\end{aligned}$$

Each parenthesized product is an element of S by closure, so applying the laws to such intermediate elements is legitimate.

Exercise 2.27: The singleton case

Let $S = \{s\}$. The only possible output of $s * s$ is s. Hence $s * s = s * s$ proves commutativity, and $(s * s) * s = s * s = s = s * (s * s)$ proves associativity. These exhaust all pairs and triples because the set has one element.

Exercise 2.28: Commutativity need not imply associativity

On $\{0, 1\}$ define $a * b = 1 - ab$ (the values are again 0 or 1). It is commutative because $ab = ba$. But $(0 * 0) * 1 = 1 * 1 = 0$, while $0 * (0 * 1) = 0 * 1 = 1$. Thus this commutative two-element operation is not associative.

Exercise 2.29: Associativity of function addition

For every real x, $((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x)$. The middle equality is associativity of real addition. Since the two functions agree at every input, they are equal, so function addition is associative.

Exercise 2.30: Commutativity of function subtraction

False. Let $f(x) = 1$ and $g(x) = 0$ for every real x. Then $(f - g)(x) = 1$ while $(g - f)(x) = -1$, so the resulting functions differ. Constant functions are allowed elements of F.

Exercise 2.31: Associativity of function subtraction

False. Let f, g, h all be the constant function 1. Then $(f - g) - h$ is the constant -1, whereas $f - (g - h)$ is the constant 1. This counterexample uses functions with exactly the required domain and codomain.

Exercise 2.32: Commutativity of function multiplication

For every real x, $(fg)(x) = f(x)g(x) = g(x)f(x) = (gf)(x)$ by commutativity of real multiplication. Pointwise agreement proves $fg = gf$ as functions. Thus the statement is true.

Exercise 2.33: Associativity of function multiplication

For every x, $((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x)$. Associativity of real multiplication gives the central equality. Thus the two functions agree on all of $\mathbb{R}$, proving the assertion.

Exercise 2.34: Commutativity of composition

False. Set $f(x) = x + 1$ and $g(x) = 2x$. Then $(f \circ g)(x) = 2x + 1$ but $(g \circ f)(x) = 2x + 2$. The order of applying functions matters, even though composition is associative.

Exercise 2.35: An arbitrary distributive law

False: choose both operations to be ordinary addition on $\mathbb{Z}$. With $a = 1, b = c = 0$, the left side $a + (b + c)$ is 1, whereas $(a + b) + (a + c)$ is 2. Distributivity is an extra relation between operations; it does not follow from their merely being binary operations.

Exercise 2.36: Closure of elements commuting with everything

Let $a, b \in H$ and $x \in S$. Reassociate and use that a, b commute with x:

$$(a * b) * x = a * (b * x) = a * (x * b) = (a * x) * b = (x * a) * b = x * (a * b).$$

Since this holds for every $x \in S$, the element $a * b$ belongs to H. No commutativity of the whole set is assumed; we only used the defining property of these two elements.

Exercise 2.37: Closure of idempotents

Take $a, b \in H$, so $a * a = a$ and $b * b = b$. Associativity and commutativity allow regrouping as

$$(a * b) * (a * b) = (a * a) * (b * b) = a * b.$$

For justification, first write $a*(b*(a*b))$, commute b past a inside, then reassociate into $(a*a)*(b*b)$. Thus $a * b$ is idempotent and lies in H.